

Azure Medallion Lakehouse Data Pipeline

A modern cloud-based data engineering and analytics pipeline built using Microsoft Azure services following the Medallion Architecture (Bronze–Silver–Gold) approach.

Overview

The Azure Medallion Lakehouse Data Pipeline is an end-to-end data engineering architecture designed to ingest, process, store, and visualize enterprise data efficiently.

The pipeline integrates multiple Azure services including:

- Azure Data Factory
- Azure Data Lake Storage Gen2
- Azure Databricks
- Azure Synapse Analytics
- Azure SQL Server
- Power BI

This architecture enables scalable data ingestion, transformation, analytics, and business intelligence reporting.

Architecture

Pipeline Flow



Architecture Components

1. Data Source Layer

The pipeline starts with on-premises or external data sources such as:

- Databases
- APIs
- ERP systems
- CSV/Excel files
- Business applications

These systems generate raw enterprise data for processing.

2. Ingestion Layer — Azure Data Factory

Azure Data Factory (ADF) is used to ingest data into Azure Data Lake Storage.

Responsibilities

- Data extraction
- Pipeline orchestration
- ETL/ELT workflow automation
- Scheduling and monitoring

Benefits

- Scalable ingestion pipelines
 - Low-code workflow management
 - Integration with multiple sources
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Medallion Architecture

The storage layer follows the Medallion Architecture pattern.

Bronze Layer — Raw Data

Stores raw and unprocessed data exactly as received.

Features

- Immutable storage
- Historical data retention
- Minimal transformation

Purpose

- Preserve original source data
 - Enable auditing and replay
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Silver Layer — Cleaned Data

Stores cleaned and transformed data.

Operations

- Data cleansing
- Deduplication
- Schema validation
- Standardization

Purpose

- Improve data quality
 - Prepare datasets for analytics
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Gold Layer — Business Ready Data

Contains curated and aggregated datasets optimized for reporting and analytics.

Operations

- KPI calculations
- Aggregations
- Business metrics generation

Purpose

- Business intelligence
 - Dashboard reporting
 - Analytical querying
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Processing Layer

Azure Databricks

Used for distributed big data processing using Apache Spark.

Responsibilities

- Data transformation
- Batch processing
- Data engineering workflows
- Large-scale analytics

Benefits

- High-performance processing
 - Scalable compute environment
 - Delta Lake integration
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Azure Synapse Analytics

Used for enterprise analytics and data warehousing.

Responsibilities

- SQL-based analytics
- Big data querying
- Data warehousing
- Reporting support

Benefits

- Unified analytics platform
 - Fast analytical queries
 - Enterprise scalability
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Serving Layer — Azure SQL Server

Processed Gold-layer data is stored in Azure SQL Server for downstream applications and reporting.

Features

- Structured relational storage
- Fast SQL querying
- Reporting optimization

Visualization Layer — Power BI

Power BI is used for interactive dashboards and business reporting.

Features

- Real-time dashboards
 - KPI tracking
 - Interactive reports
 - Business insights
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End-to-End Workflow

Step 1 — Data Ingestion

Data is extracted from on-premises systems using Azure Data Factory and loaded into Azure Data Lake Storage.

Step 2 — Data Processing

Databricks and Synapse process the data through:

- Bronze Layer
- Silver Layer
- Gold Layer

using Delta Lake architecture.

Step 3 — Data Serving

Curated business-ready datasets are loaded into Azure SQL Server.

Step 4 — Data Visualization

Power BI connects to Azure SQL Server and generates dashboards and reports.

Key Features

- Cloud-native architecture
 - Scalable data processing
 - Medallion data architecture
 - Distributed computing with Spark
 - Enterprise analytics support
 - Centralized data governance
 - BI-ready datasets
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Advantages

- Better data quality and governance
 - Automated orchestration
 - High-performance processing
 - Reliable and consistent storage
 - Enterprise-scale analytics
 - Interactive visualization
 - Fast reporting access
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Technologies Used

- Azure Data Factory — Data ingestion and orchestration
 - Azure Data Lake Storage Gen2 — Scalable cloud storage
 - Delta Lake — Data reliability and versioning
 - Azure Databricks — Data processing
 - Azure Synapse Analytics — Enterprise analytics
 - Azure SQL Server — Structured serving layer
 - Power BI — Reporting and visualization
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Use Cases

- Enterprise Data Warehousing
- Business Intelligence Platforms
- Real-Time Analytics
- Data Lakehouse Solutions
- Financial Reporting

- Customer Analytics
 - Operational Dashboards
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Conclusion

The Azure Medallion Lakehouse Data Pipeline provides a scalable and efficient framework for enterprise data engineering and analytics. By combining Azure cloud services with the Medallion Architecture approach, the system transforms raw enterprise data into actionable business insights through automated ingestion, processing, storage, and visualization workflows.